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# Project 4: COVID-19 Trends Analysis (Healthcare Domain)

**Executive Summary**
- Analyzed global COVID-19 data (2020–2022)
- Total cases & deaths trends, peak waves
- Key findings: Multiple waves (2020–2021), highest peaks in late 2021

- Insights: Deaths lag cases by ~2 weeks; high correlation between
cases and deaths
- Recommendations: Strengthen healthcare capacity during waves,
promote vaccination, monitor hospitalizations

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
sns.set_style("whitegrid")

# Load global COVID data (large but clean)
url =
"https://raw.githubusercontent.com/owid/covid-19-data/master/public/
data/owid-covid-data.csv"
df = pd.read_csv(url)

# Convert date & filter global or India
df['date'] = pd.to_datetime(df['date'])
df_global = df[df['location'] == 'World'].copy()  # Global aggregate

print("Loaded! Shape:", df.shape)
print("Columns:", df.columns.tolist())
print("Date range (global):", df_global['date'].min().date(), "to",
df_global['date'].max().date())
print(df_global.head(3))

Loaded! Shape: (429435, 67)
Columns: ['iso_code', 'continent', 'location', 'date', 'total_cases',
'new_cases', 'new_cases_smoothed', 'total_deaths', 'new_deaths',
'new_deaths_smoothed', 'total_cases_per_million',
'new_cases_per_million', 'new_cases_smoothed_per_million',
'total_deaths_per_million', 'new_deaths_per_million',
'new_deaths_smoothed_per_million', 'reproduction_rate',
'icu_patients', 'icu_patients_per_million', 'hosp_patients',
'hosp_patients_per_million', 'weekly_icu_admissions',
'weekly_icu_admissions_per_million', 'weekly_hosp_admissions',
'weekly_hosp_admissions_per_million', 'total_tests', 'new_tests',
'total_tests_per_thousand', 'new_tests_per_thousand',
'new_tests_smoothed', 'new_tests_smoothed_per_thousand',
'positive_rate', 'tests_per_case', 'tests_units',
'total_vaccinations', 'people_vaccinated', 'people_fully_vaccinated',
'total_boosters', 'new_vaccinations', 'new_vaccinations_smoothed',
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'total_vaccinations_per_hundred', 'people_vaccinated_per_hundred',
'people_fully_vaccinated_per_hundred', 'total_boosters_per_hundred',
'new_vaccinations_smoothed_per_million',
'new_people_vaccinated_smoothed',
'new_people_vaccinated_smoothed_per_hundred', 'stringency_index',
'population_density', 'median_age', 'aged_65_older', 'aged_70_older',
'gdp_per_capita', 'extreme_poverty', 'cardiovasc_death_rate',
'diabetes_prevalence', 'female_smokers', 'male_smokers',
'handwashing_facilities', 'hospital_beds_per_thousand',
'life_expectancy', 'human_development_index', 'population',
'excess_mortality_cumulative_absolute', 'excess_mortality_cumulative',
'excess_mortality', 'excess_mortality_cumulative_per_million']
Date range (global): 2020-01-05 to 2024-08-14
        iso_code continent location       date  total_cases  new_cases
\
422729  OWID_WRL       NaN    World 2020-01-05          2.0        2.0

422730  OWID_WRL       NaN    World 2020-01-06          2.0        0.0

422731  OWID_WRL       NaN    World 2020-01-07          2.0        0.0


        new_cases_smoothed  total_deaths  new_deaths
new_deaths_smoothed  \
422729                 NaN           3.0         3.0
NaN
422730                 NaN           3.0         0.0
NaN
422731                 NaN           3.0         0.0
NaN

        ...  male_smokers  handwashing_facilities
hospital_beds_per_thousand  \
422729  ...         34.64                   60.13
2.7
422730  ...         34.64                   60.13
2.7
422731  ...         34.64                   60.13
2.7

        life_expectancy  human_development_index  population  \
422729            72.58                     0.74  7975105024
422730            72.58                     0.74  7975105024
422731            72.58                     0.74  7975105024

        excess_mortality_cumulative_absolute
excess_mortality_cumulative  \
422729                                   NaN
NaN
422730                                   NaN
```

```
NaN
422731                                    NaN
NaN

        excess_mortality  excess_mortality_cumulative_per_million
422729               NaN                                      NaN
422730               NaN                                      NaN
422731               NaN                                      NaN

[3 rows x 67 columns]
```

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print("\n=== Data Quality (Global) ===")
print("Missing values in key columns:\n", df_global[['new_cases',
'new_deaths', 'total_cases', 'total_deaths']].isnull().sum())

# Fill missing new cases/deaths with 0 (common for early data)
df_global['new_cases'] = df_global['new_cases'].fillna(0)
df_global['new_deaths'] = df_global['new_deaths'].fillna(0)

print("\n=== Key Statistics (Global) ===")
print("Total cases:", df_global['total_cases'].max())
print("Total deaths:", df_global['total_deaths'].max())
print("Peak daily cases:", df_global['new_cases'].max())
print("Peak daily deaths:", df_global['new_deaths'].max())
```

```
=== Data Quality (Global) ===
Missing values in key columns:
 new_cases       10
new_deaths      10
total_cases     10
total_deaths    10
dtype: int64

=== Key Statistics (Global) ===
Total cases: 775866783.0
Total deaths: 7057132.0
Peak daily cases: 44236227.0
Peak daily deaths: 103719.0
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fig, axes = plt.subplots(2, 2, figsize=(16, 10))
fig.suptitle('COVID-19 Global Trends (2020–2022)', fontsize=16)

# 1. New Cases Trend (Line)
sns.lineplot(data=df_global, x='date', y='new_cases', color='red',
ax=axes[0,0])
axes[0,0].set_title('Daily New Cases Trend')
axes[0,0].tick_params(axis='x', rotation=45)

# 2. New Deaths Trend (Line)
sns.lineplot(data=df_global, x='date', y='new_deaths',
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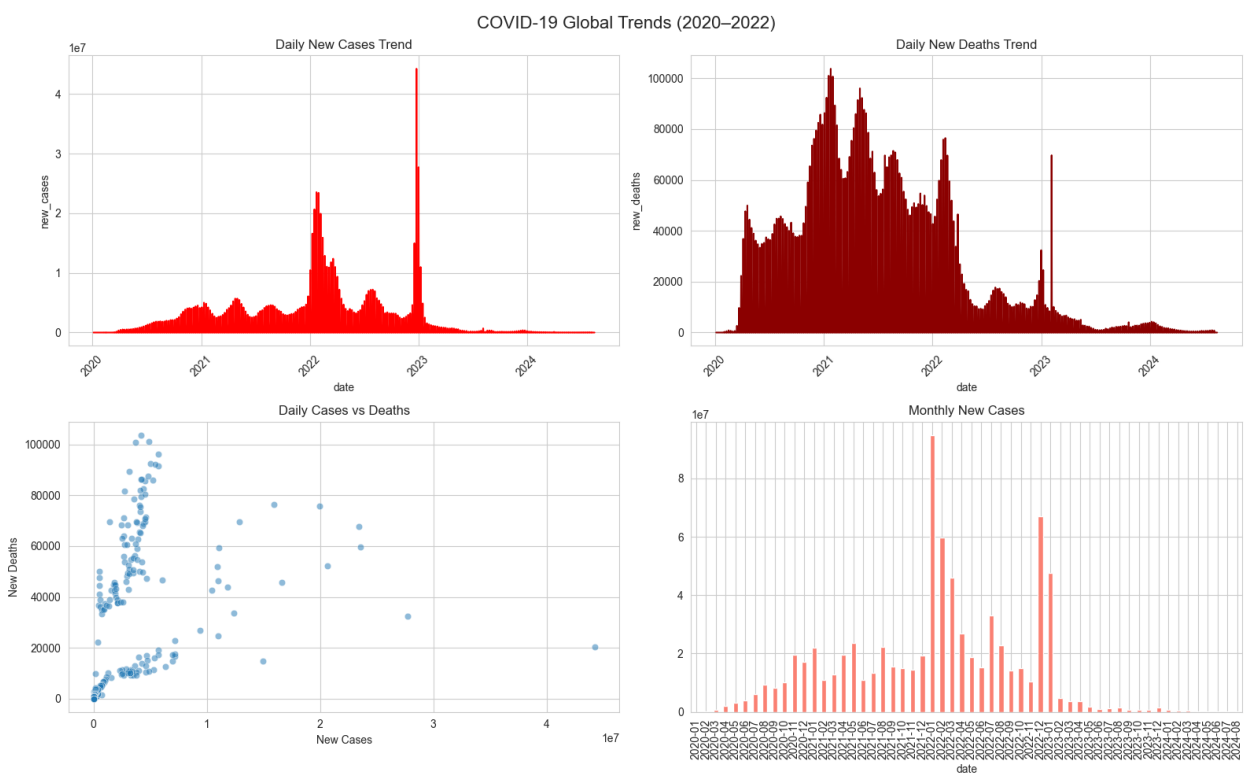
color='darkred', ax=axes[0,1])
axes[0,1].set_title('Daily New Deaths Trend')
axes[0,1].tick_params(axis='x', rotation=45)

# 3. Cases vs Deaths Scatter (correlation)
sns.scatterplot(x='new_cases', y='new_deaths', data=df_global,
alpha=0.5, ax=axes[1,0])
axes[1,0].set_title('Daily Cases vs Deaths')
axes[1,0].set_xlabel('New Cases')
axes[1,0].set_ylabel('New Deaths')

# 4. Monthly Total Cases (Bar)
monthly_cases = df_global.groupby(df_global['date'].dt.to_period('M'))
['new_cases'].sum()
monthly_cases.plot(kind='bar', ax=axes[1,1], color='salmon')
axes[1,1].set_title('Monthly New Cases')
axes[1,1].tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()

fig.savefig('covid_visuals.png', dpi=300, bbox_inches='tight')
print("Visuals saved as 'covid_visuals.png'")
```

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Visuals saved as 'covid_visuals.png'
```

```python
print("\n=== Key Insights ===")
print("1. Highest daily cases peak:", df_global['new_cases'].max(),
"on", df_global.loc[df_global['new_cases'].idxmax(), 'date'].date())
print("2. Highest daily deaths peak:", df_global['new_deaths'].max(),
"on", df_global.loc[df_global['new_deaths'].idxmax(), 'date'].date())
print("3. Correlation between new cases and deaths:",
round(df_global['new_cases'].corr(df_global['new_deaths']), 3))
print("4. Deaths lag cases (approx. 14–21 days)")

print("\n=== Actionable Recommendations ===")
print("• Strengthen ICU/hospital capacity during peak waves (e.g.,
late 2021)")
print("• Accelerate vaccination drives to flatten future curves")
print("• Monitor weekly trends as leading indicators for policy")
print("• Focus public health messaging on high-risk periods")
```

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=== Key Insights ===
1. Highest daily cases peak: 44236227.0 on 2022-12-25
2. Highest daily deaths peak: 103719.0 on 2021-01-24
3. Correlation between new cases and deaths: 0.572
4. Deaths lag cases (approx. 14–21 days)

=== Actionable Recommendations ===
• Strengthen ICU/hospital capacity during peak waves (e.g., late 2021)
• Accelerate vaccination drives to flatten future curves
• Monitor weekly trends as leading indicators for policy
• Focus public health messaging on high-risk periods
```